

JIACHENG LIANG

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RESEARCH INTEREST

My research focuses on **large language models** and **trustworthy AI**, spanning LLM post-training and alignment, reinforcement learning for retrieval and recommendation, retrieval-augmented generation, agentic LLM systems, and robust model deployment. I build end-to-end ML pipelines for improving the reliability, safety, and real-world effectiveness of AI systems across applications such as recommendation, LLM serving, and AI security.

EDUCATION

Stony Brook University & Penn State University

Ph.D. Candidate in Computer Science, GPA: 4.0/4.0

Stony Brook, NY

Sep 2022 – Dec 2026 (Expected)

University of Electronic Science and Technology of China

B.Eng. in Software Engineering (International Elite Class)

Chengdu, China

Sep 2018 – Jun 2022

EXPERIENCE

Prime Video & Amazon MGM Studios

Applied Scientist Intern

Seattle, WA

Jun 2026 – Present

- **Reinforcement Learning for Multi-Intent Recommendation.** Built *Intent-Q*: decomposes a user's coexisting intents (long-term preference, session context, exploration, time-sensitive trends) and trains a single $\leq 4B$ policy with **GRPO** to emit one retrieval query per intent over a **frozen dense retriever**. Designed a continuous **global-rank reward** that stays informative in large-catalog cold-start, where truncated NDCG@k is identically zero and gives no gradient. (Pub. *Intent-Q*)

Amazon AGI Foundation, Nova Responsible AI Team

Applied Scientist Intern

Boston, MA

May 2025 – Aug 2025

- **Adaptive Red-Teaming for RLHF (ARES).** Identified *systemic weaknesses* in **RLHF** pipelines where the policy and the **reward model** fail in tandem, making an imperfect RM a single point of failure that policy-level red-teaming never surfaces. Built a "Safety Mentor" that composes semantically coherent adversarial prompts from structured components (topics, personas, tactics, goals) with paired malicious/safe responses, probing policy and RM at once, and drove a two-stage repair that first fine-tunes the RM to detect harmful content and then optimizes the policy against the improved RM, substantially strengthening safety robustness on adversarial benchmarks while preserving model capability. (Pub. *ARES*, *ACL '26*)

SELECTED PUBLICATIONS

Advisor: Prof. Ting Wang (ALPS-Lab). Entries are first / co-first authored unless tagged **co-author**. **★ Top** marks top-tier venues.

Reinforcement Learning & Recommendation

Intent-Q: Reinforced Query Generation for Decomposed Multi-Intent Recommendation

Amazon Prime Video · Under review

GRPO-trained $\leq 4B$ policy emits one retrieval query per coexisting user intent over a **frozen dense retriever**; a continuous **global-rank reward** restores gradients in large-catalog cold-start. +38% Complement NDCG@10 on Amazon ESCI.

LLM Post-Training & Alignment

ARES: Adaptive Red-Teaming and End-to-End Repair of Policy-Reward Systems

Amazon AGI · ACL '26 ★ Top 

Adaptive **red-teaming** exposes joint policy/reward failures in **RLHF**; end-to-end repair closes alignment gaps.

RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models

EMNLP '26 ★ Top 

Routing-aware safety alignment for **MoE** LLMs: bi-level optimization robustifies experts and routers.

Data to Defense: The Role of Curation in Customizing LLMs Against Jailbreaking Attacks

EMNLP '25 ★ Top 

End-to-end **data curation** across **LLM fine-tuning** stages, reaching a 100% safe-response rate against adversarial customization.

LLM Agents & Memory

AgentLAB: Benchmarking LLM Agents Against Long-Horizon Attacks

ICML '26 ★ Top co-author 

Extensible benchmark for **LLM agents** under **long-horizon** adversarial attacks, uncovering vulnerabilities beyond single-turn defenses.

MAGE: Safeguarding LLM Agents against Long-Horizon Threats via Shadow Memory

ACM CCS '26 ★ Top co-author 

Agentic **shadow memory** distills safety-critical context to detect and mitigate **long-horizon threats** before unsafe actions execute.

Retrieval-Augmented Generation (RAG)

GraphRAG under Fire

IEEE S&P '26 (Oakland) ★ Top 

First poisoning attack on **GraphRAG**: entity-relation graph corruption hijacks downstream **retrieval-augmented generation**.

LLM Inference & Serving Efficiency

Robust Semantic Cache for LLM Serving

Under review

Semantic cache for LLM serving that indexes request + **speculatively-decoded** tokens to resist cache poisoning while improving retrieval relevance.

Dynamic Token Reweighting for Robust Vision-Language Models

CVPR '26 ★ Top co-author 

Dynamic KV-cache reweighting for VLMs: the first inference-time multimodal defense, with no retraining required.

RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction

ICLR '25 ★ Top co-author 

KV-cache eviction drops low-importance tokens to neutralize adversarial inputs without hurting benign performance.

Reasoning Models & Red-Teaming

AutoRAN: Automated Hijacking of Safety Reasoning in Large Reasoning Models

ACL '26 ★ Top 

Automated attack on **large reasoning models (LRMs)**, exposing how **chain-of-thought** transparency becomes an exploitable surface.

Benchmarking & Evaluation

WaterPark: A Robustness Assessment of Language Model Watermarking

EMNLP '25 ★ Top 

Systematization of **LLM watermarking**: standardized robustness eval, a novel attack, and corresponding defenses.

Model Extraction Attacks Revisited (MEBench)

AsiaCCS '24

Open-source benchmark for **model extraction** across real **MLaaS** APIs, unifying attacks, metrics, and victim models.